

ARCO 17 SW, ID, USA

WMO: 725790

Lat: 43.4621N

Lon: 113.5560W

Elev: 1804

StdP: 81.44

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 04126

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-16.0	-22.0	0.7	-17.2	-19.7	0.8	-14.3	11.4	-1.4	9.9	-1.5	1.1	N/A	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	31.7	13.0	30.3	12.5	28.9	12.2	14.5	26.2	13.8	26.0	13.2	25.5	4.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.0	10.2	16.1	9.5	9.2	16.1	8.2	8.4	16.1	46.6	26.3	44.4	25.8	42.7	25.4	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.2	8.1	7.1	DB	-21.5	34.4	2.4	1.1	-23.2	35.2	-24.6	35.8	-26.0	36.4	-27.7	37.2	
			WB	-21.4	15.9	2.9	0.8	-23.5	16.4	-25.2	16.9	-26.8	17.3	-28.9	17.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-7.1	-5.3	0.1	5.4	10.3	15.6	21.6	20.2	14.8	7.0	-0.4	-6.5
	DBStd	10.83	4.63	4.74	4.68	4.22	4.46	4.54	2.95	3.26	4.71	4.95	5.08	4.92
	HDD10.0	2433	528	430	309	149	52	8	0	0	15	118	312	511
	HDD18.3	4591	787	663	567	389	249	103	8	19	122	352	562	769
	CDD10.0	1111	0	0	1	11	62	176	361	318	157	25	0	0
	CDD18.3	227	0	0	0	0	1	22	111	79	14	1	0	0
	CDH23.3	2641	0	0	0	1	24	285	1273	875	178	4	0	0
	CDH26.7	864	0	0	0	0	1	76	478	279	29	1	0	0
Wind	WSAvg	2.6	2.0	2.6	3.0	3.1	3.0	3.0	2.8	2.7	2.6	2.5	2.4	2.1
Precipitation	PrecAvg	283	32	23	21	28	35	30	13	14	15	21	19	32
	PrecMax	420	61	71	70	51	83	87	29	30	30	53	52	87
	PrecMin	190	3	1	0	8	3	3	1	2	2	3	3	7
	PrecStd	71	18	18	18	11	25	24	9	10	8	16	13	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	8.1	15.7	21.3	26.0	32.0	33.8	32.8	29.8	23.0	14.0	5.7
		MCWB	0.9	3.4	5.7	8.5	10.5	13.9	14.0	12.9	12.2	10.6	6.4	2.1
	2%	DB	1.9	5.0	12.4	18.4	23.5	28.7	32.3	31.1	27.2	19.9	11.4	3.6
		MCWB	-0.3	1.4	4.6	7.1	9.9	12.4	13.2	12.5	11.3	8.8	4.9	1.8
	5%	DB	0.4	3.0	9.5	15.8	21.3	26.7	30.9	29.8	25.5	17.2	8.8	1.9
		MCWB	-1.2	0.2	3.2	6.0	9.4	11.6	12.8	12.2	10.7	7.6	3.7	0.3
	10%	DB	-0.9	1.4	6.8	13.3	19.0	24.5	29.6	28.2	23.6	15.0	6.6	0.4
		MCWB	-2.2	-0.6	2.2	5.1	8.5	10.9	12.4	11.9	10.1	6.8	2.5	-1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	4.0	6.6	9.4	11.9	15.2	15.5	15.1	15.1	10.8	7.1	3.2
		MCDB	3.4	7.7	14.0	20.3	20.1	29.8	27.8	23.8	21.6	19.8	13.9	4.7
	2%	WB	0.6	2.7	5.1	7.9	10.7	13.8	14.5	14.1	13.2	10.0	5.5	2.0
		MCDB	2.1	5.7	10.9	16.7	20.0	27.2	27.4	24.3	22.0	17.1	10.9	3.4
	5%	WB	-0.4	1.4	3.9	6.7	9.8	12.6	13.9	13.4	12.1	8.8	4.2	0.6
		MCDB	0.6	3.5	8.8	14.8	18.9	24.8	27.2	24.7	21.9	15.6	8.3	1.9
	10%	WB	-1.6	0.2	2.9	5.5	9.0	11.7	13.4	12.8	11.1	7.5	3.0	-0.9
		MCDB	-0.5	2.0	6.6	12.2	17.4	23.2	26.6	24.9	21.0	13.7	6.4	0.2
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.8	8.0	10.8	11.7	13.0	14.2	14.0	12.9	10.8	8.5	6.6
		MCDBR	7.7	7.9	11.6	14.2	14.2	15.4	15.5	15.4	14.8	13.1	10.7	7.2
		MCWBR	6.4	5.8	6.4	7.1	6.2	5.8	5.7	5.8	6.0	6.2	6.1	6.0
	5% WB	MCDBR	7.3	7.9	10.1	12.9	12.9	14.6	14.3	13.7	12.5	11.5	10.5	7.0
		MCWBR	6.2	5.7	5.8	6.8	5.8	5.5	5.3	5.3	5.2	5.9	6.2	5.9
Clear-Sky Solar Irradiance	taub	0.229	0.238	0.255	0.283	0.296	0.298	0.307	0.307	0.286	0.264	0.245	0.230	
	taud	2.453	2.489	2.574	2.521	2.532	2.557	2.547	2.520	2.590	2.630	2.611	2.516	
	Ebn at Noon	937	986	1003	988	975	970	957	949	953	943	917	908	
	Edh at Noon	70	81	86	99	101	99	99	98	84	71	60	60	
All-Sky Solar Radiation	RadAvg	1.75	2.68	4.02	5.37	6.40	7.46	7.54	6.51	5.13	3.46	2.13	1.45	
	RadStd	0.20	0.35	0.37	0.38	0.60	0.56	0.39	0.37	0.34	0.23	0.16	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page